

Total No. of Questions : 8]

SEAT No. :

PE2737

[Total No. of Pages : 2

[6583]-318

T.E. (Computer Engineering) (Honors)
STATISTICS AND MACHINE LEARNING
(2019 Pattern) (Semester - VI) (310503)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Explain [9]

- i) Eigen Value
- ii) Eigen Vector

Find Eigen Values of a 2 by 2 matrix $A = \begin{bmatrix} 2 & -1 \\ -4 & 5 \end{bmatrix}$.

- b) Write formulae for Jacobian matrix and Hessian Matrix. Find the Jacobian matrix at the point (1,2) of the following function:
 $f(x, y) = (x^4 + 3y^2x, 5y^2 - 2xy + 1)$. [9]

OR

Q2) a) Write short note on [9]

- i) Chain Rule and its Significance
- ii) Partial Derivatives

- b) What is linear equation? What are the different methods to solve system of linear equation? Explain with suitable example. [9]

Q3) a) Explain different ways in which Netflix uses Machine Learning Algorithms. [9]

- b) Explain in detail Supervised and Unsupervised learning in Machine Learning. [8]

OR

P.T.O.

Q4) a) What is Machine Learning, with the help of an example and diagram explain Reinforcement Learning. [9]

b) List and explain various [8]

i) Applications of Machine Learning

ii) Issues in machine learning

Q5) a) What is Regression. Explain different types of Regressions in detail. [9]

b) Explain gradient descent algorithm. Elaborate, how it can be used to get minimum cost function. [9]

OR

Q6) a) Write short note on [9]

i) Model training

ii) Model Evaluation

iii) Confusion Matrix

b) Explain Data Normalization and its necessity. Explain in detail any one data normalization technique. [9]

Q7) a) What are the Advantages, Disadvantages and Applications of Naïve Bayes Model. [9]

b) What is a decision tree and how to construct it? Explain the various decision tree terminologies. [8]

OR

Q8) a) Explain with example hypothesis space search in decision tree learning? [9]

b) For the given data set where R stands for Rainy, O stands for Overcast and S stands for Sunny using Naïve bayes algorithm check if the player would play or not if the weather is sunny. [8]

Dataset														
Weather	R	S	O	O	S	R	S	O	R	S	S	R	O	O
Play	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	Yes	No	Yes	Yes

x

x

x